

Year 9 Digital Technologies

Unit 1 — Websites 101

St Brendan's College (Yeppoon), 2026



Unit overview	Cohort and/or class considerations
Start week: Term 1 Week 1 Finish week: Term 1 Week 5 In this unit, students will explore the foundational technologies behind the internet and how websites function. They will examine how digital systems use hardware and software to manage and secure access to data across networks. Through guided investigation and hands-on tasks, students will learn how websites are constructed by separating content, structure, and presentation using HTML and CSS. They will also be introduced to simple data compression techniques and understand their importance in efficient web communication. This unit builds the knowledge and skills required for designing and analysing secure, well-structured, and efficient digital content online.	The Years 9–10 cohort consists of a diverse group of boys with a wide range of digital competencies, from students with strong technical skills and personal interest in IT, to those who require further support to confidently engage with digital systems. Many students come from regional or rural backgrounds, where access to structured digital learning opportunities may have been limited. The program is designed to build on prior knowledge while introducing more advanced and applied skills relevant to future study and employment. The curriculum supports the school's priority to equip students with transferable digital skills, critical thinking abilities, and an understanding of ethical technology use in a rapidly changing world.

Adjustments	
Student initials/identifier	
Year level/band against which the student is to be assessed	Band 9–10
Category of need	
Adjustment/s and/or key information	
Review of adjustments	

Assessment 1	
Year 9	
Title	Web Exam
Description	Students will complete a written exam that assesses their understanding of how websites function and how data is managed in networked digital systems. The exam will include questions on the roles of hardware and software in securing and controlling access to data, the structure of web documents using HTML and CSS, and basic data compression techniques.
Technique	Examination
Mode	Written
Conditions	• up to 50 minutes plus 10 minutes planning time • under supervised conditions • short response up to 100 words per item, up to 600 words for the task • extended response 200–300 words per item, up to 600 words for the task
Timing	Week 5
Achievement standard	By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation . They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data , and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.
Moderation	A sample of student portfolios representing a range of achievement levels will be reviewed by another Digital Technologies teacher, who will provide feedback on the consistency and fairness of the grading. Any necessary adjustments will be made to ensure alignment with the assessment criteria and standards.

Content descriptions
Knowledge and understanding
Digital systems investigate how hardware and software manage, control and secure access to data in networked digital systems (AC9TDI10K01)

Content descriptions

Knowledge and understanding

Data representation

represent documents online as content (text), structure (markup) and presentation (styling) and explain why such representations are important (AC9TDI10K02)

investigate simple data compression techniques (AC9TDI10K03)

General capabilities

Critical and creative thinking

- Inquiring
 - Develop questions
 - Identify, process and evaluate information
- Generating
 - Create possibilities
 - Consider alternatives
 - Put ideas into action
- Analysing
 - Interpret concepts and problems
 - Draw conclusions and provide reasons
 - Evaluate actions and outcomes
- Reflecting
 - Transfer knowledge

Digital literacy

- Practising digital safety and wellbeing
 - Manage online safety
 - Manage digital privacy and identity
 - Manage digital wellbeing
- Investigating
 - Locate information
 - Acquire and collate data
 - Interpret data
- Creating and exchanging
 - Plan
 - Create, communicate and collaborate
 - Respect intellectual property
- Managing and operating
 - Select and operate tools

Literacy

- Reading and viewing
 - Understanding texts
- Writing

Teaching and learning sequence

Week	Planning details	Theory	Practical	Assessment
1	Key teaching and learning experiences	Students investigate how the internet functions as a network of digital systems, focusing on the roles of hardware (servers, routers, client devices) and software (browsers, web servers) in managing and transmitting data. Key terminology such as client-server model, IP address, and DNS is introduced. Students develop an understanding of	Students explore how the web works through guided simulations and demonstrations (e.g. request-response cycle). They use browser developer tools to inspect simple websites and identify basic structural elements. Introductory activities include navigating file structures	Introduction to assessment requirements and expectations for the Web Exam. Diagnostic questioning to assess prior knowledge. Short-response practice focusing on defining key terms and explaining simple processes (e.g. "How does a browser retrieve a webpage?").

Teaching and learning sequence				
		how data is requested, transferred, and displayed in a web context.	and preparing a workspace for web development.	
	Adjustments			
	Resources	• Teacher slides on “How the Internet Works” • Short video explaining client-server model • Browser developer tools (Chrome/ Edge) • Introductory worksheets on internet concepts		
2	Key teaching and learning experiences	Students are introduced to how web content is structured using HTML. The concept of representing information as content and structure is explicitly taught. Students learn about common HTML elements (headings, paragraphs, images, links) and the purpose of semantic structure.	Students begin writing basic HTML documents using a code editor. They create a simple webpage including headings, paragraphs, and images. Emphasis is placed on correct syntax, file structure, and readability of code.	Ongoing formative assessment through teacher observation and feedback on HTML tasks. Short-response questions focused on explaining the role of HTML and identifying elements within code snippets.
	Adjustments			
	Resources	• HTML reference guide (basic tags) • Teacher demonstration files • Online HTML editor or VS Code • Structured practice worksheets		
3	Key teaching and learning experiences	Students explore the concept of presentation using CSS and how it separates design from structure. Key concepts include selectors, properties, and values, as well as the importance of consistency and maintainability in web design.	Students extend their existing HTML pages by applying CSS styling. Tasks include changing colours, fonts, spacing, and layout. Students experiment with linking external stylesheets and modifying visual presentation without altering content.	Formative assessment through completion of a styled webpage task. Students respond to short-answer questions explaining the difference between HTML and CSS and why separation is important. Teacher provides feedback on both code and explanations.

Teaching and learning sequence				
	Adjustments			
	Resources	• CSS reference guide (basic properties) • Sample styled websites • Teacher demonstration of CSS implementation • Practice exercises (styling challenges)		
4	Key teaching and learning experiences	Students investigate simple data compression techniques and their importance in web communication. Concepts such as file size, load time, and efficiency are explored. Students examine examples of compressed vs uncompressed data and discuss the impact on user experience.	Students apply optimisation techniques to their webpages, such as resizing images and reducing file sizes. They compare performance differences between optimised and non-optimised files. Activities may include using online tools to compress images and analyse file sizes.	Short-response and extended-response practice questions focusing on explaining data compression and its importance. Retrieval practice covering all topics to date (digital systems, HTML, CSS). Teacher provides targeted feedback in preparation for the exam.
	Adjustments			
	Resources	• Teacher slides on data compression • Image compression tools (e.g. TinyPNG or similar) • Example files for comparison • Exam-style question sets		
5	Key teaching and learning experiences	Structured revision of key concepts: digital systems, HTML (content and structure), CSS (presentation), and data compression. Students consolidate their understanding through guided review and discussion.	Revision activities including quizzes, concept mapping, and peer discussion. Students complete practice exam questions under timed conditions to build familiarity with assessment requirements.	Completion of the Web Exam under supervised conditions (as per assessment specifications). Final review and clarification of expectations prior to the exam.
	Adjustments			
	Resources	• Revision slides and summary sheets • Practice exam papers • Key terminology lists • Past/sample questions		

This document may contain confidential personal information, and must be handled in accordance with applicable privacy law.

Copyright in the template of this document belongs to the QCAA and ACARA and is licensed [Creative Commons Attribution 4.0](#). The QCAA requires no attribution. For the latest information and additional terms of use of the ACARA's content, please check the [Australian Curriculum website](#) and its [copyright notice](#). Content provided by St Brendan's College (Yeppoon) belongs to the school. Contact the school to obtain permissions to reuse its content, where required.